GEN-AI HANDS-ON 1

NAME:ANNSHUL SHETTY

SRN:PES2UG23CS079

Observation:

This notebook shows how combines hugging face transformers with NLP tools like NLTK. During tokenization and POS tagging there were errors as recent NTLK updates that split the resources into language specific files. Downloading the required new datasets fixed the issue, allowing the notebook to continue smoothly from basic preprocessing to advanced NLP tasks like NER, summarization, and question answering.

SEED VALUE:42

DISTILL MODEL:

Let's see how the smaller model performs.

```
# Initialize the pipeline with the specific model
fast_generator = pipeline('text-generation', model='distilgpt2')

# Generate text
output_fast = fast_generator(prompt, max_length=50, num_return_sequences=1)
print(output_fast[0]['generated_text'])
```

```
Device set to use cpu
Truncation was not explicitly activated but `max_length` is provided a specific value, please use `
Setting `pad_token_id` to `eos_token_id`:50256 for open-end generation.
Both `max_new_tokens` (=256) and `max_length`(=50) seem to have been set. `max_new_tokens` will tak
Generative AI is a revolutionary technology that is designed to work with existing AI systems. It h


The research team led by Professor Daniel Kranz, from the University of California, Berkeley, has d
The research team developed the program to learn how to use the AI to improve the performance of th
The research team developed the program to learn how to use the AI to improve the performance of th
```

STANDARD MODEL:

Now let's try the standard model.

```
[78]
19s
smart_generator = pipeline('text-generation', model='gpt2')

output_smart = smart_generator(prompt, max_length=50, num_return_sequences=1)
print(output_smart[0]['generated_text'])
```

```
Device set to use cpu
Truncation was not explicitly activated but `max_length` is provided a specific value, please use `
Setting `pad_token_id` to `eos_token_id`:50256 for open-end generation.
Both `max_new_tokens` (=256) and `max_length`(=50) seem to have been set. `max_new_tokens` will tak
Generative AI is a revolutionary technology that allows users to build AI that can help solve compl

The AI is a model of human intelligence, and has many aspects that are similar to artificial intell

It is the main driving force behind the new Artificial Intelligence, and the AI is very important t

The AI is designed to be scalable and adaptable to different environments. It can be used to solve

The new AI is designed to work out problems that need to be solved in a way that is
```

SEED VALUE:66

DISTILL MODEL:

```
[52]
12s
# Initialize the pipeline with the specific model
fast_generator = pipeline('text-generation', model='distilgpt2')

# Generate text
output_fast = fast_generator(prompt, max_length=50, num_return_sequences=1)
print(output_fast[0]['generated_text'])
```

```
Device set to use cpu
Truncation was not explicitly activated but `max_length` is provided a specific value, please use
Setting `pad_token_id` to `eos_token_id`:50256 for open-end generation.
Both `max_new_tokens` (=256) and `max_length`(=50) seem to have been set. `max_new_tokens` will ta
Generative AI is a revolutionary technology that can be harnessed by the human brain to create a s
```

STANDARD MODEL:

```
53]
✓ 18s
smart_generator = pipeline('text-generation', model='gpt2')

output_smart = smart_generator(prompt, max_length=50, num_return_sequences=1)
print(output_smart[0]['generated_text'])
```

```
Device set to use cpu
Truncation was not explicitly activated but `max_length` is provided a specific value, please use `
Setting `pad_token_id` to `eos_token_id`:50256 for open-end generation.
Both `max_new_tokens` (=256) and `max_length`(=50) seem to have been set. `max_new_tokens` will tak
Generative AI is a revolutionary technology that transforms human perception into a fully sentient

This technology is called "AI for Everything," and it is a very different kind of artificial intell

Advertisement

That's why AI is so important to us. We think machines are going to do more and more things that hu

What is more, we've built so many of these artificial intelligence systems over the last decade tha

You see a lot of people talking about how we can automate things. It's true that this technology is

Advertisement

As a whole, we're going to be able to automate more and more of what humans have done. We're going
```

Q & A:

```
[92]
✓ 6s
questions = [
    "What is the fundamental innovation of the Transformer?",
    "What are the risks of using Generative AI?"
]

for q in questions:
    res = qa_pipeline(question=q, context=text[:5000])
    print(f"\nQ: {q}")
    print(f"A: {res['answer']}")
```

```
Q: What is the fundamental innovation of the Transformer?
A: to identify hidden patterns, structures, and relationships within the data

Q: What are the risks of using Generative AI?
A: data privacy, intellectual property, and academic integrity
```

SEED VAUES:

When I used seed 42, the model generated the same response every time because the randomness followed a fixed path, making the output predictable and easy to reproduce. When I switched to seed 66, the model didn't become smarter or worse it simply made different random choices while generating text. Because language models choose the next word from several valid possibilities, even a small change in the starting seed led to different words, sentence flow, and overall tone. Both outputs were correct and meaningful they just reflected different decisions In simple terms, the prompt and model stayed the same, but changing the seed changed how the model expressed its answer.

STANDARD MODEL VS DISTILL MODEL:

The standard model is larger and usually gives more detailed and coherent answers because it understands context better. The distilled model is a smaller, faster version that uses less memory but may repeat ideas or lose focus sometimes. Simply put, the standard model focuses on quality, while the distilled model focuses on speed and efficiency.